

Ctrl-MC Software Stack Setup + Getting Started



This document describes the software stack and method used to set up the development environment for the Ctrl-MC project. This guide is for a PC setup with Windows 11, but many of the instructions work for other platforms as well.

If you don't want to create your own project, but just want to be able to modify and compile the existing Ctrl-MC source code, you should still make sure all the software mentioned below is installed and the extensions in VS Code are set up. You can skip the instructions on how to set up a new project, instead just download the source code from [GitHub](#). Use VS Code and select "File" + "Open Folder..." + select the folder for the code you want to work with.

Overview

The microcontrollers used for the Ctrl-MC system are from the STM32 family of microcontrollers (mcu). For the very first prototypes made in 2025/2026 the STM32G431 is used.

Software to be used is **STM32CubeMX** for generating the initial code, **STM32CubeCLT** for third-party integrated dev environments, and **Microsoft VS Code** with STM32 extensions for code editing, for uploading the code to mcu and for debugging. In addition **CMake** must be installed for

STM32CubeMX

The software: **STM32CubeMX** is used to generate the initial code needed to interact with a STM32 microcontroller. The software can be downloaded for free from [st.com](#) website. You will have to create an account to download, but the software is completely free.

- STM32CubeMX: <https://www.st.com/en/development-tools/stm32cubemx.html>

For creating a new development project, start the application and select "Start My project from ST Board" from ST Board"

Start My project from ST Board

ACCESS TO BOARD SELECTOR

Select the correct board from the tab "Board Selector" if using a STM dev board. If using a 3rd party board, like the "WeAct Studio" dev board used for the first prototype, identify the mcu ID, for example: "STM32G431CBU6". Select the tab "MCU/MPU Selector" and use the field "Commercial Part Number" to find the correct MCU.

When the correct board or MCU is found, select: Start Project.

Start Project

Now the tab "Pinout & Configuration" should be selected, and the mcu should be shown visualized similarly to the picture to the right with all its pins. Each pin to be used must be selected and configured.

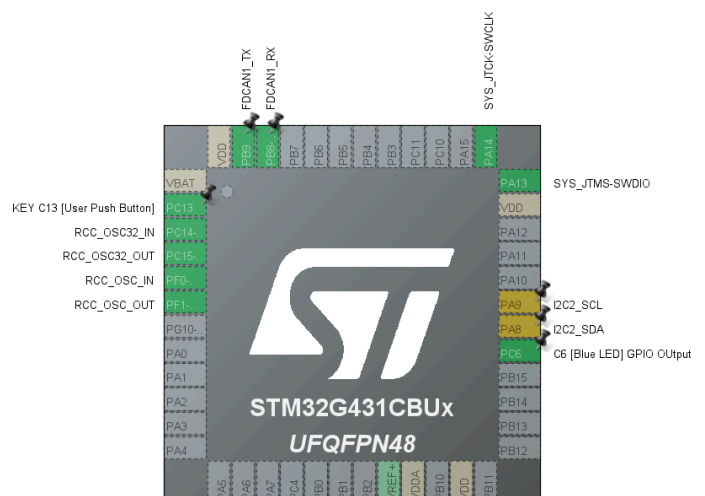
The next tab is the "Clock Configuration", this needs also to be configured to set the speed for mcu.

When configuration is done, select the third tab: "Project Manager". Remember to set "CMake" as build system, this what Ctrl-MC project users. Set at least this:

- Project name (no spaces in the name)
- Project location where you want the files saved
- Toolchain / IDE select: **CMake**

The config should be done, select the button that says: generate code.

GENERATE CODE



STM32CubeCLT

STM32CubeCLT is a toolset for third-party integrated development environment (IDE) providers, allowing the use of STMicroelectronics proprietary tools within their own IDE frameworks.

- STM32CubeCLT: <https://www.st.com/en/development-tools/stm32cubecolt.html>

Follow the user manual install instructions from the [documents tab from the download page](#).

Microsoft Visual Studio Code

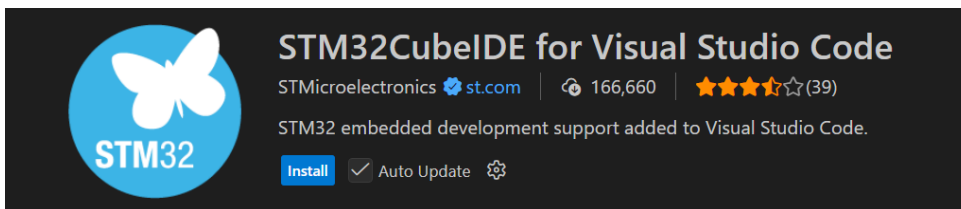
The preferred code editor is MS VS code. It has an extension that makes it convenient to upload the code to the mcu and for debugging code while running it.

- Microsoft Visual Studio Code: <https://code.visualstudio.com/download>

After install select the extension selector in the left sidebar, and search for:



- STM32Cube for Visual Studio Code



CMake

CMake is a cross-platform, open-source build system generator that uses configuration files to automate the creation of platform-specific build scripts (like Makefiles or Visual Studio projects) for managing the software compilation process.

CMake is needed to be installed for compiling the STM32CubeIDE project and for uploading it to the MCU since this is what was set in STM32CubeMX as the Toolchain / IDE parameter.

Check here for more info, how to download and install CMake: <https://cmake.org/download>

On Windows it's possible to install it from an elevated command prompt like this: `choco install cmake` You might need to refresh path environment variables or just restart the PC after install.

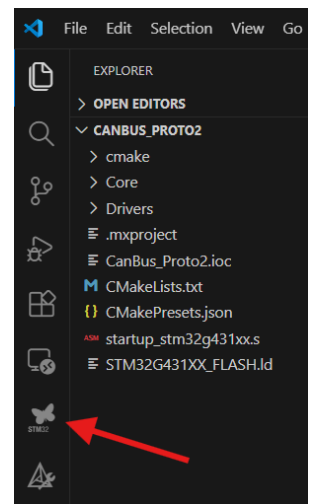
Getting started with a new STM32 project in Visual Studio

Setting up your own project is a smart way to understand the fundamentals. It's highly recommended to do this before diving into others' code. Here is a quick guide for how to get started.

This assumes all the software on the previous pages is set up and a new project is created using STM32CubeMX. Open VS Code and navigate to the root folder of the project by selecting "File" + "Open Folder..." and select the folder.

The picture to the right shows an example folder structure of a newly created project named: "CANBUS_PROTO2".

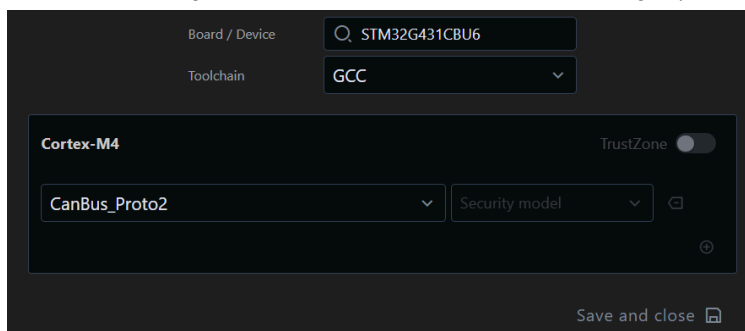
Navigate to the STM32 extension in the left navigation bar and select "Setup STM32Cube project(s)" (double click).



Fill in the project setup by selecting the same board/device as used in the STM32CubeMX configuration, and use the GCC toolchain. The picture below shows a config using the "WeAct Studio" development board with a STM32G431 mcu on it.



Press the "Configure" button and then select the "Debug" option, and then "Save and close"



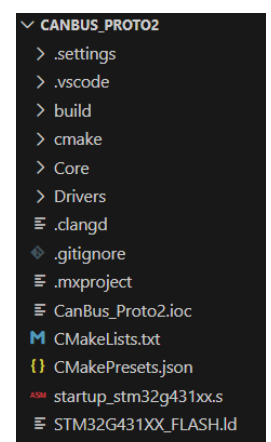
When this is done, additional folders (.settings, .vscode and build) will be created for the project.

Go to the folder: **Core + Src + main.c** to start building your first code, remember to always add your custom code in between of comments like:

```
/* USER CODE BEGIN 2 */

// Blink 5 times rapidly to indicate startup
for (int i = 0; i < 10; i++)
{
    HAL_GPIO_TogglePin(C6_BLUE_LED_GPIO_Port, C6_BLUE_LED_Pin);
    HAL_Delay(50);
}
HAL_GPIO_WritePin(C6_BLUE_LED_GPIO_Port, C6_BLUE_LED_Pin, GPIO_PIN_RESET);

/* USER CODE END 2 */
```



This code should now be possible to build and upload to the microcontroller. Good Luck!

Revisions

Version 1 - Created by KI Hestad: 2025-12-16